PREMIUM

Full-Length Mock Test

Mock 01

Geology and Geophysics

GATE GG

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20% | **Moderate 33%** | **Hard 47%**

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	55	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		**60**		**100**	

Marking Scheme

- Correct MCQ: +1 mark (or +2 marks as specified per question)
- Wrong MCQ: -0.33 marks
- Correct MSQ: +2 marks
- Wrong MSQ (partial or no match): -0.33 marks
- Correct NAT: +1 mark (or +2 marks as specified per question)
- No answer / Unattempted: 0 marks
- NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Physical Geology — Minerals]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.2 [1 Mark] [EASY] [Mineralogy — Physical Properties]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.3 [1 Mark] [EASY] [Geophysics — Magnetic]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.4 [1 Mark] [EASY] [Physical Geology — Minerals]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.5 [1 Mark] [EASY] [Mineralogy — Crystal Systems]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.6 [1 Mark] [EASY] [Geophysics — Electrical Methods]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.7 [1 Mark] [EASY] [Physical Geology — Minerals]

Hardness on Mohs scale for diamond:

A) 1

B) 5

C) 10

D) 7

Q.8 [1 Mark] [EASY] [Mineralogy — Optical Properties]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.9 [1 Mark] [EASY] [Geophysics — Magnetic]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.10 [1 Mark] [EASY] [Physical Geology — Weathering]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.11 [1 Mark] [EASY] [Mineralogy — Physical Properties]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.12 [1 Mark] [EASY] [Geophysics — Seismic Methods]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.13 [2 Marks] [MODERATE] [Physical Geology — Folds/Faults]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.14 [2 Marks] [MODERATE] [Mineralogy — Physical Properties]

P-wave velocity in rocks is typically:

A) 1-2 km/s

B) 3-6 km/s

C) 8-10 km/s

D) 10-12 km/s

Q.15 [2 Marks] [MODERATE] [Geophysics — Electrical Methods]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.16 [2 Marks] [MODERATE] [Physical Geology — Weathering]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.17 [2 Marks] [MODERATE] [Mineralogy — Crystal Systems]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.18 [2 Marks] [MODERATE] [Geophysics — Electrical Methods]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.19 [2 Marks] [MODERATE] [Physical Geology — Folds/Faults]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.20 [2 Marks] [MODERATE] [Mineralogy — Optical Properties]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.21 [2 Marks] [MODERATE] [Geophysics — Magnetic]

Hardness on Mohs scale for diamond:

A) 1

B) 5

C) 10

D) 7

Q.22 [2 Marks] [MODERATE] [Physical Geology — Weathering]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.23 [2 Marks] [MODERATE] [Mineralogy — Physical Properties]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.24 [2 Marks] [MODERATE] [Geophysics — Electrical Methods]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.25 [2 Marks] [MODERATE] [Physical Geology — Folds/Faults]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.26 [2 Marks] [MODERATE] [Mineralogy — Optical Properties]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.27 [2 Marks] [MODERATE] [Geophysics — Electrical Methods]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.28 [2 Marks] [MODERATE] [Physical Geology — Rocks]

P-wave velocity in rocks is typically:

A) 1-2 km/s

B) 3-6 km/s

C) 8-10 km/s

D) 10-12 km/s

Q.29 [2 Marks] [MODERATE] [Mineralogy — Optical Properties]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.30 [2 Marks] [MODERATE] [Geophysics — Electrical Methods]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.31 [2 Marks] [MODERATE] [Physical Geology — Minerals]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.32 [2 Marks] [MODERATE] [Mineralogy — Optical Properties]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.33 [2 Marks] [HARD] [Geophysics — Gravity]

Hardness on Mohs scale for diamond:

A) 1
B) 5
C) 10
D) 7

Q.34 [2 Marks] [HARD] [Physical Geology — Weathering]

P-wave velocity in rocks is typically:

A) 1-2 km/s
B) 3-6 km/s
C) 8-10 km/s
D) 10-12 km/s

Q.35 [2 Marks] [HARD] [Mineralogy — Physical Properties]

Hardness on Mohs scale for diamond:

A) 1

B) 5

C) 10

D) 7

Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Geophysics — Magnetic]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.37 [2 Marks] [HARD] [Physical Geology — Folds/Faults]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.38 [2 Marks] [HARD] [Mineralogy — Physical Properties]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.39 [2 Marks] [HARD] [Geophysics — Electrical Methods]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Q.40 [2 Marks] [HARD] [Physical Geology — Minerals]

Which statements are correct?

A) Statement A
B) Statement B
C) Statement C
D) Statement D
E) Statement E

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Geophysics — Magnetic]

Standard calculation using basic formula:

Answer: ____________________

Q.52 [2 Marks] [HARD] [Physical Geology — Minerals]

Standard calculation using basic formula:

Answer: ____________________

Q.53 [2 Marks] [HARD] [Mineralogy — Crystal Systems]

Standard calculation using basic formula:

Answer: ____________________

Q.54 [2 Marks] [HARD] [Geophysics — Electrical Methods]

Standard calculation using basic formula:

Answer: ____________________

Q.55 [2 Marks] [HARD] [Physical Geology — Folds/Faults]

Standard calculation using basic formula:

Answer: ____________________

Q.56 [2 Marks] [HARD] [Mineralogy — Crystal Systems]

Standard calculation using basic formula:

Answer: ____________________

Q.57 [2 Marks] [HARD] [Geophysics — Magnetic]

Standard calculation using basic formula:

Answer: ____________________

Q.58 [2 Marks] [HARD] [Physical Geology — Folds/Faults]

Standard calculation using basic formula:

Answer: ________________________

Q.59 [2 Marks] [HARD] [Mineralogy — Physical Properties]

Standard calculation using basic formula:

Answer: ____________________________

Q.60 [2 Marks] [HARD] [Geophysics — Seismic Methods]

Standard calculation using basic formula:

Answer: ____________________________

Answer Key

Verify your answers after completing the full paper.

Q.1:	C	**Q.2:**	B	**Q.3:**	C
Q.4:	B	**Q.5:**	C	**Q.6:**	B
Q.7:	C	**Q.8:**	B	**Q.9:**	C
Q.10:	B	**Q.11:**	C	**Q.12:**	B
Q.13:	C	**Q.14:**	B	**Q.15:**	C
Q.16:	B	**Q.17:**	C	**Q.18:**	B
Q.19:	C	**Q.20:**	B	**Q.21:**	C
Q.22:	B	**Q.23:**	C	**Q.24:**	B
Q.25:	C	**Q.26:**	B	**Q.27:**	C
Q.28:	B	**Q.29:**	C	**Q.30:**	B
Q.31:	C	**Q.32:**	B	**Q.33:**	C
Q.34:	B	**Q.35:**	C	**Q.36:**	AC
Q.37:	AC	**Q.38:**	AC	**Q.39:**	AC
Q.40:	AC	**Q.41:**	AC	**Q.42:**	AC
Q.43:	AC	**Q.44:**	AC	**Q.45:**	AC
Q.46:	AC	**Q.47:**	AC	**Q.48:**	AC
Q.49:	AC	**Q.50:**	AC	**Q.51:**	10
Q.52:	10	**Q.53:**	10	**Q.54:**	10
Q.55:	10	**Q.56:**	10	**Q.57:**	10
Q.58:	10	**Q.59:**	10	**Q.60:**	10

Detailed Solutions

Question 1
[Physical Geology — Minerals]

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 2
[Mineralogy — Physical Properties]

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 3
[Geophysics — Magnetic]

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 4
[Physical Geology — Minerals]

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 5
[Mineralogy — Crystal Systems]

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 6
[Geophysics — Electrical Methods]

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 7
[Physical Geology — Minerals]

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 8
[Mineralogy — Optical Properties]

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 9
[Geophysics — Magnetic]

Hardness on Mohs scale for diamond:

Answer:

(C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 10 *[Physical Geology — Weathering]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 11 *[Mineralogy — Physical Properties]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 12 *[Geophysics — Seismic Methods]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 13 *[Physical Geology — Folds/Faults]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 14 *[Mineralogy — Physical Properties]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 15 *[Geophysics — Electrical Methods]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 16 *[Physical Geology — Weathering]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 17 *[Mineralogy — Crystal Systems]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 18 *[Geophysics — Electrical Methods]*

P-wave velocity in rocks is typically:

Answer:

(B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 19 *[Physical Geology — Folds/Faults]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 20 *[Mineralogy — Optical Properties]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 21 *[Geophysics — Magnetic]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 22 *[Physical Geology — Weathering]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 23 *[Mineralogy — Physical Properties]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 24 *[Geophysics — Electrical Methods]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 25 *[Physical Geology — Folds/Faults]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 26 *[Mineralogy — Optical Properties]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 27 *[Geophysics — Electrical Methods]*

Hardness on Mohs scale for diamond:

Answer:

(C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 28 *[Physical Geology — Rocks]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 29 *[Mineralogy — Optical Properties]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 30 *[Geophysics — Electrical Methods]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 31 *[Physical Geology — Minerals]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 32 *[Mineralogy — Optical Properties]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 33 *[Geophysics — Gravity]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 34 *[Physical Geology — Weathering]*

P-wave velocity in rocks is typically:

Answer: (B)

P-wave velocity in crustal rocks: 3-6 km/s.

Question 35 *[Mineralogy — Physical Properties]*

Hardness on Mohs scale for diamond:

Answer: (C)

Diamond is hardest mineral with Mohs hardness = 10.

Question 36 *[Geophysics — Magnetic]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 37 *[Physical Geology — Folds/Faults]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 *[Mineralogy — Physical Properties]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 *[Geophysics — Electrical Methods]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 *[Physical Geology — Minerals]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 *[Mineralogy — Optical Properties]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 *[Geophysics — Seismic Methods]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 *[Physical Geology — Rocks]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 *[Mineralogy — Crystal Systems]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 *[Geophysics — Electrical Methods]*

Which statements are correct?

Answer:

(A, C)

Based on theoretical analysis of the topic.

Question 46 *[Physical Geology — Folds/Faults]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 *[Mineralogy — Optical Properties]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 *[Geophysics — Electrical Methods]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 *[Physical Geology — Weathering]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 *[Mineralogy — Optical Properties]*

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 *[Geophysics — Magnetic]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 52 *[Physical Geology — Minerals]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 53 *[Mineralogy — Crystal Systems]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 54 *[Geophysics — Electrical Methods]*

Standard calculation using basic formula:

Answer:

(10)

Apply the appropriate formula for the given data.

Question 55 *[Physical Geology — Folds/Faults]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 56 *[Mineralogy — Crystal Systems]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 57 *[Geophysics — Magnetic]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 58 *[Physical Geology — Folds/Faults]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 59 *[Mineralogy — Physical Properties]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 60 *[Geophysics — Seismic Methods]*

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.
